ELECTIVE COURSE-2.3

URA731: ARTIFICIAL INTELLIGENCE IN PRODUCTION				
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	2	0	2	3.0

Course Objective: Learn how Artificial Intelligence and machine learning is being applied to industrial production, train a process that takes over the fill level monitoring of the chutes on the sorting inline system.

Syllabus

Artificial intelligence: Introduction to Artificial intelligence, Artificial intelligence in everyday life, Artificial intelligence in production system, fundamentals of Artificial intelligence Machine learning and deep learning, supervised learning, unsupervised learning, Reinforcement learning, neural nets, Tenure track, image classification, object localization and object detection in a nutshell, supervised, unsupervised and reinforcement learning in a nutshell, Application scenario: Optical quality inspection, optical quality inspection via deep learning, neural networks, biological neural networks, Artificial neural networks.

ML Framework: Functionality of model software for image classification and image identification, Introduction to image classification, Image classification and Optimization of Training parameters, Handling Training Data, input and output encoding of the neural net in image classification, limitations of optical quality inspection using ML-based image classification,

Machine learning vision: A practical introduction to AI/ML, image classification vs object detection, naïve implementation of object detection, modern ML-based object detection, modern ML-based object localization, anomaly detection, saving images, annotating images, Training the neural net, Testing the neural net, re-training the neural net, testing and re-training the neural net, configuration of files, receiving via mqtt, memory bottlenecks, publishing images via mqtt, application module camera inspection, switching between PLC and ML mode, Introduction to python, Machine learning concepts, applied statics, neural language processing, face detection, object detection, tensorflow and neural network, motion analysis and object detection, skill based Application with Sorting Station operation and

Machine Vision Learning.

Laboratory Work

1. Introduction to Artificial intelligence, explore the terms Machine learning, artificial intelligence, deep learning.
2. Introduction to image classification software tool
3. Image classification and optimization of Training parameters
4. Handling Training data, Understanding the working principles of Sorting station.
5. Calculate the productivity and efficiency of a production system.
6. Program to monitor process control of the station and monitor the process control
7. Remote access to Raspberry Pi
8. Teaching Monitor operation in sorting in line station.
9. the setup and process sequence of the MPS 400 Sorting Inline.

Course Learning Outcomes (CLOs)

The students will be able to:

1. understand the basics of Machine learning, Artificial intelligence, Deep learning, supervised learning with the classification method.
2. implement machine learning process to a new situation by formulating new requirements.
3. develop optimized neural net by extending existing training data and interpret the learning progress of a training using the Accuracy and Loss vs. Epochs charts.
4. apply remote access to the Raspberry Pi to view all saved data.
5. understand the purpose of a test data set and how it is statistically evaluated in the image classification application.

Text Books

1. Rich E., Knight K. and Nair B. S., Artificial Intelligence, Tata McGraw Hills (2009) 3rd ed

Reference Books & Websites

1. Object detection with AI, LX
2. Machine Vision,LX

Hardware Reference:

1. Sorting inline Station
2. Machine learning vision.

Evaluation Scheme:

SI. No	Evaluation Elements	Weightage (%)
1	MST	35
2	EST	35
3	Sessional (May include Assignments/Projects/Tutorials/Quiz/Lab	30

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.